HDOC-70

Digital Root Using Recursion and Static Variable

Problem Statement

Write a recursive function in C to calculate the digital root of a number. Use a static variable to count the number of recursive calls.

Algorithm / Logic

1. Create a recursive function digitalRoot() and declare a static variable callCount inside it.

2. Increment callCount every time digitalRoot() is called.

3. If the number is a single digit (0 to 9), it is the digital root. Store the total recursive-call count and return the number.

4. Otherwise, calculate the sum of all digits of the current number.

5. Call digitalRoot() recursively with this digit sum.

6. In main(), read the number, call the recursive function, and display both the digital root and the number of recursive calls.

C Program

```
#include <stdio.h>

int totalCalls = 0;

int digitalRoot(int n)
{
    static int callCount = 0;
    int sum = 0;

    callCount++;

    if (n < 10)
    {
        totalCalls = callCount;
        return n;
    }

    while (n > 0)
    {
        sum += n % 10;
        n /= 10;
    }

    return digitalRoot(sum);
}

int main()
{
    int num, root;
```

```
    printf("Enter a positive integer: ");
    scanf("%d", &num);

    if (num < 0)
    {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    root = digitalRoot(num);

    printf("Digital root = %d\n", root);
    printf("Number of recursive calls = %d\n", totalCalls);

    return 0;
}
```

Explanation

The function first counts its invocation using the static variable callCount. A static variable retains its value between recursive calls. If the current number has more than one digit, the program finds the sum of its digits and recursively calls digitalRoot() with that sum. Recursion stops when the value becomes a single digit. That final digit is the digital root.

For 9875: 9 + 8 + 7 + 5 = 29, then 2 + 9 = 11, then 1 + 1 = 2. Therefore, the digital root is 2. The function is invoked four times: for 9875, 29, 11, and 2.

Test Cases

Input	Reduction	Digital Root	Recursive Calls
9875	9875 -> 29 -> 11 -> 2	2	4
12345	12345 -> 15 -> 6	6	3
999	999 -> 27 -> 9	9	3
7	7	7	1
0	0	0	1

Sample Output

```
Enter a positive integer: 9875
Digital root = 2
Number of recursive calls = 4
```

Result

The program successfully calculates the digital root of a given positive integer using recursion and uses a static variable to count the number of recursive function calls.